



Memphis Zoo Custodial System

Screens, workflow, and cost control

A clear walkthrough of how schedules, scans, guest reports, events, reminders, Memphis Messenger, and Memphis AI work together during the day.

01 See the work

Managers can see scan activity, completion status, guest issues, and schedules from the screens they already use.

02 Guide the day

Custodians get simple phone workflows: assigned areas, scan prompts, event notices, due soon reminders, and issue notifications.

03 Keep proof simple

NFC/QR activity creates a time-stamped trail without asking employees to manage spreadsheets or paperwork.

04 Control cost

The design uses low-cost phones, NFC tags, kiosk licenses, GitHub Pages, Supabase, Render, and zoo Wi-Fi instead of a broad commercial software license.

Important framing

The system does not replace manager inspections. It helps managers see which areas are ready to check, so inspections can happen at the right time.

Document style

This PDF is designed for print or PC viewing and includes full application screenshots.

How the System Works

The idea is simple: when something is scanned, reported, or overdue, the right screen updates and the right person is notified.

1. Area ownership

Schedules assign custodians to restroom, exhibit, and service areas for the day.

2. Work starts at the location

The custodian uses the phone workflow and NFC/QR scan points to identify the real location.

3. Keeps working offline

If Wi-Fi is weak, the phone keeps working and queues the action.

4. Updates the dashboard

When the connection returns, queued activity syncs and dashboard views refresh.

5. Messenger notifications

Guest issues, event notices, due soon locations, and overdue locations create Memphis Messenger alerts.

6. Manager inspection

Managers use the screens to see what is ready to inspect. The software supports the inspection; it does not replace it.

The strength is that everything is connected. Instead of separate tools for schedules, scans, guest issues, events, messages, and follow-up, the zoo gets one workflow built around the actual day.

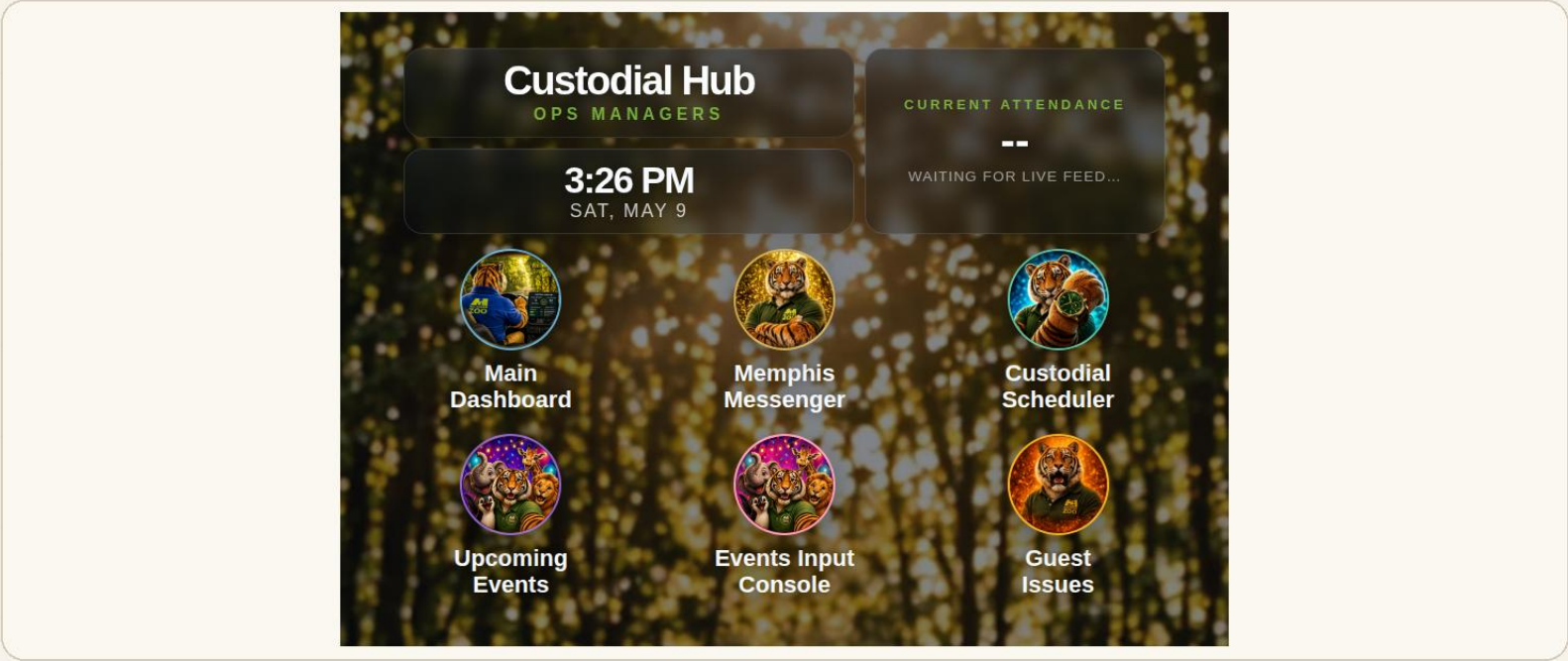
WHAT CHANGES OPERATIONALLY

- Less guessing about which areas are ready for inspection.
- Less dependence on radio calls and memory when mornings get busy.
- Clearer response path when a guest reports a cleanliness issue.
- Event information reaches the people assigned to the affected area.
- Due-soon and overdue reminders help prevent missed work.

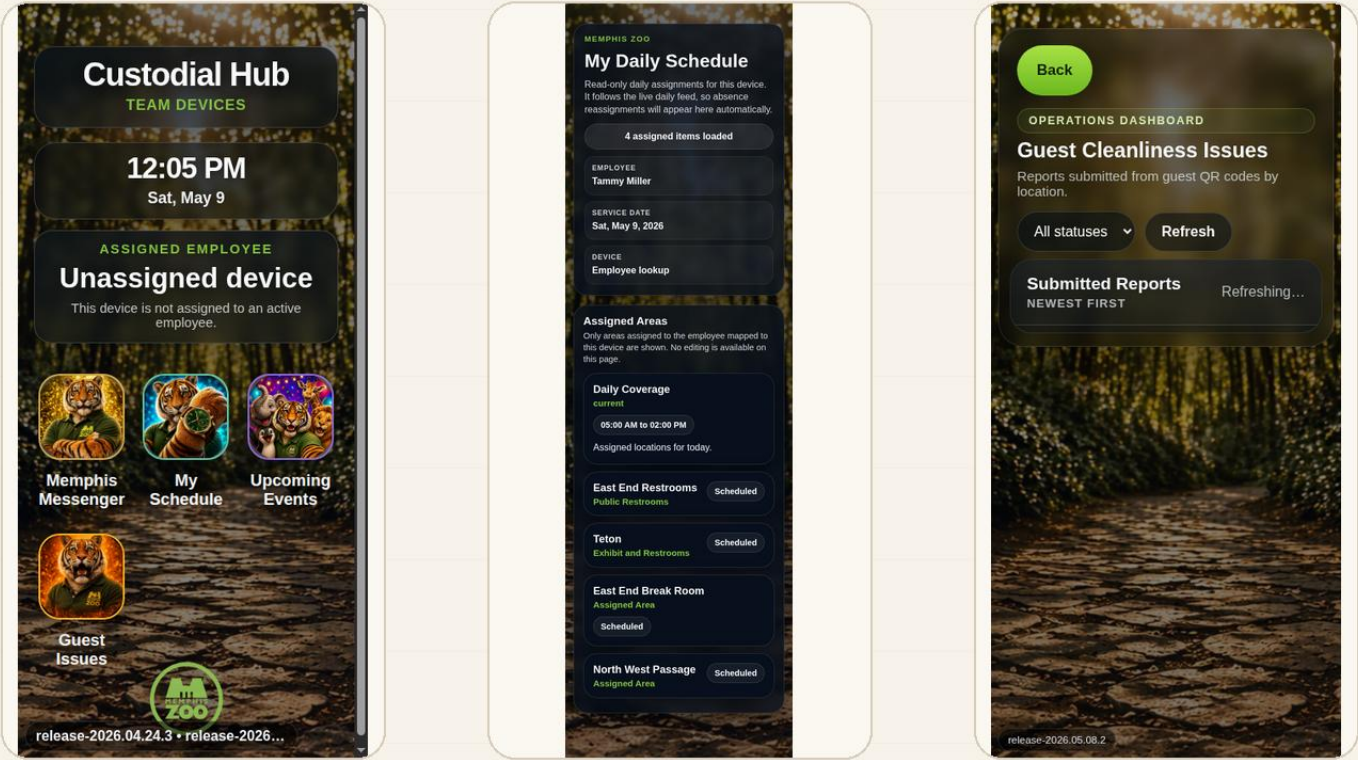
Ops Manager Hub + Custodian Phone Views

03

The manager starts from one PC hub while custodians use simple phone screens in the field.



Ops Manager Hub — cropped just below the action icons



Employee Hub

Tammy Miller — My Schedule

Guest Issues View

HOW THE TWO SIDES WORK TOGETHER

- Managers use the hub to open dashboard, schedules, Messenger, events, and guest issues from one place.
- Custodians use phone screens for assigned areas, scan work, messages, events, and guest issue follow-up.
- The same information flows between both sides: schedule ownership, guest reports, due-soon reminders, overdue locations, and inspection timing.

Operations Dashboard

Fuller PC view showing cleaned areas, scan activity, open issues, and inspection readiness.

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OPERATIONS DASHBOARD

TODAY
5:54 PM
Sat, May 9, 2026
CURRENT ATTENDANCE
8,124

Restrooms Overdue
1

Restrooms Not Cleaned
0

Open Tickets
4

Exhibits Overdue
1

Exhibits Not Cleaned
0

Restroom Status Report

LIVE OPERATIONAL STATE

LOCATION	STATUS	TIME	TIME TO CLEAN	SERVICES PERFORMED	EMPLOYEE NAME
Teton Trek Restrooms	OVERDUE	Yesterday 4:58 PM	22 min	Full cleaning and inspection completed	Tammy Miller
China Restrooms	DUE SOON	Today 8:16 AM	16 min	Floors swept and mopped, Soap restocked, Trash emptied and receptacles cleaned	Marcus Reed
Front Entrance Restrooms	CLEANING IN PROGRESS	—	12 min	—	Dana Cole
Aquarium Restrooms	OKAY	Today 9:42 AM	18 min	Toilets and urinals cleaned, Sinks and counters cleaned, Paper towels restocked, Final inspection completed before scanning the location tag	Tammy Miller

Exhibit Status Report

LIVE OPERATIONAL STATE

LOCATION	STATUS	TIME	TIME TO CLEAN	SERVICES PERFORMED	EMPLOYEE NAME
Cat Country	OVERDUE	Yesterday 3:40 PM	27 min	Floors swept or vacuumed, including corners, Walls, signage, switch plates, and baseboards spot-cleaned	Riley Brooks
Aquarium Exhibit	DUE SOON	Today 7:35 AM	19 min	Entrance and exit doors cleaned, Exterior glass cleaned	Jordan Lee
Primate Canyon	CLEANING IN PROGRESS	—	9 min	—	Avery Smith
Teton Trek Exhibit	OKAY	Today 9:55 AM	24 min	Floors swept or vacuumed, including corners, Interior glass cleaned, Final inspection completed before scanning the location tag	Tammy Miller

Open Tickets

MAINTENANCE ITEMS AWAITING RESOLUTION

LOCATION	TIMESTAMP	ISSUE	REPORTED BY	ACTION
Tropical Birds	04/27/2026 09:02 PM	Floor damage observed	Karen Robinson	Close
Cathouse Cafe Men's Restroom	04/27/2026 09:01 PM	Floor drain backing up	Alijah Collins	Close
West Admin Upstairs Women's Restroom	04/22/2026 07:47 PM	Toilet continuously flushing or water running	Kathy Phelps	Close
Expo Men's Restroom	04/21/2026 12:55 AM	Soap dispenser damaged or missing	Daniel Morgan	Close

Dashboard refreshes automatically every 30 seconds.

Documentation capture: status chips use dashboard rendering; open tickets are live system rows

release-2026.04.25.3

What this screen shows

Managers can see status states, scan activity, live open tickets, and which areas are ready for inspection.

Memphis Zoo Custodial System • normal page PDF • May 2026

Daily Schedule and Absence Reassignment

Fuller PC view showing schedule coverage, absences, week readiness, and workload weighting.

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Custodial Schedule

Daily ops scheduler, ownership board, absence handling, and workload balancing.

SERVICE DATE05/09/2026

CLOSE TIME6:00 PM

Save Close

Load

Generate Missing

Generate Week

Generate

GROUPS24

Daily ownership rows

COVERED8

No open segments

OPEN SEGMENTS0

Needs manual coverage

ASSIGNED EMPLOYEES8

Unique owners today

WARNINGS16

Overloads or gaps

CLOSING TIME6:00 PM

Current configured close

Loaded 24 groups for 2026-05-09.

Week Readiness

See which days are already built before staff start asking Memphis.

0 / 7 ready

Auto week fill checked 7 generated day(s) from 2026-05-09.

0 ready, 7 missing, 0 total assignments across the visible window.

2026-05-09

Missing data

Roster: 0 • Assignments: 0

Tap to load this day

2026-05-10

Missing data

Roster: 0 • Assignments: 0

Tap to load this day

2026-05-11

Missing data

Roster: 0 • Assignments: 0

Tap to load this day

2026-05-12

Missing data

Roster: 0 • Assignments: 0

Tap to load this day

2026-05-13

Missing data

Roster: 0 • Assignments: 0

Tap to load this day

2026-05-14

Missing data

Roster: 0 • Assignments: 0

Tap to load this day

2026-05-15

Missing data

Roster: 0 • Assignments: 0

Tap to load this day

Daily Area Ownership

Compact group board with segment ownership, open coverage, and problem states.

24 groups

AllOpen onlyOverloadedCovered

Aquarium

AQUARIUM

Aquarium

Kathy Phelps

06:00 AM to 03:00 PM

Load points: 15

Saturday: full weekend-style static coverage.

Michael McWright

03:00 PM to 06:00 PM

Load points: 15

Saturday: full weekend-style static coverage.

Covered

Bonobos Restrooms

BONOBOS_RESTROOMS

Bonobos Men's Restroom, Bonobos Women's Restroom

Daniel Morgan

07:00 AM to 04:00 PM

Load points: 160

Saturday: full weekend-style static coverage. Daniel primary area correction.

Michael McWright

04:00 PM to 06:00 PM

Load points: 160

Saturday: full weekend-style static coverage.

Overloaded

Breezeway Restrooms

BREEZEWAY_RESTROOMS

Breezeway Men's Restroom, Breezeway Women's Restroom

Karen Robinson

05:00 AM to 02:00 PM

Load points: 90

ASSIGNED

Michael McWright

04:00 PM to 06:00 PM

Load points: 90

Saturday: full weekend-style static coverage.

Overloaded

Cathouse Cafe Restrooms

CATHOUSE_CAFE_RESTROOMS

Cathouse Cafe Men's Restroom, Cathouse Cafe Women's Restroom

Alijah Collins

07:00 AM to 04:00 PM

Load points: 140

Saturday: full weekend-style static coverage.

Michael McWright

04:00 PM to 06:00 PM

Load points: 140

Saturday: full weekend-style static coverage.

Overloaded

China

CHINA

China, China Men's Restroom, China Women's Restroom

Sherita Wilbon

08:00 AM to 05:00 PM

Load points: 64

Scored proximity rebuild: moved China to Sherita to improve Saturday balance.

Michael McWright

04:00 PM to 06:00 PM

Load points: 64

Saturday: full weekend-style static coverage.

Overloaded

Ops Actions

Mark absences, preview impact, then publish live changes.

0 selected

Absent custodians

Alijah Collins

Daniel Morgan

Karen Robinson

Kathy Phelps

Kinnaye Peete

Markiesha Warren

Preview Absences

Apply Absences

Select absent custodians, preview changes, or publish them to the live day schedule.

Workload Settings

Difficulty and priority run from 1 to 10. These settings feed load points used by balancing and absence reassignment.

45 locations

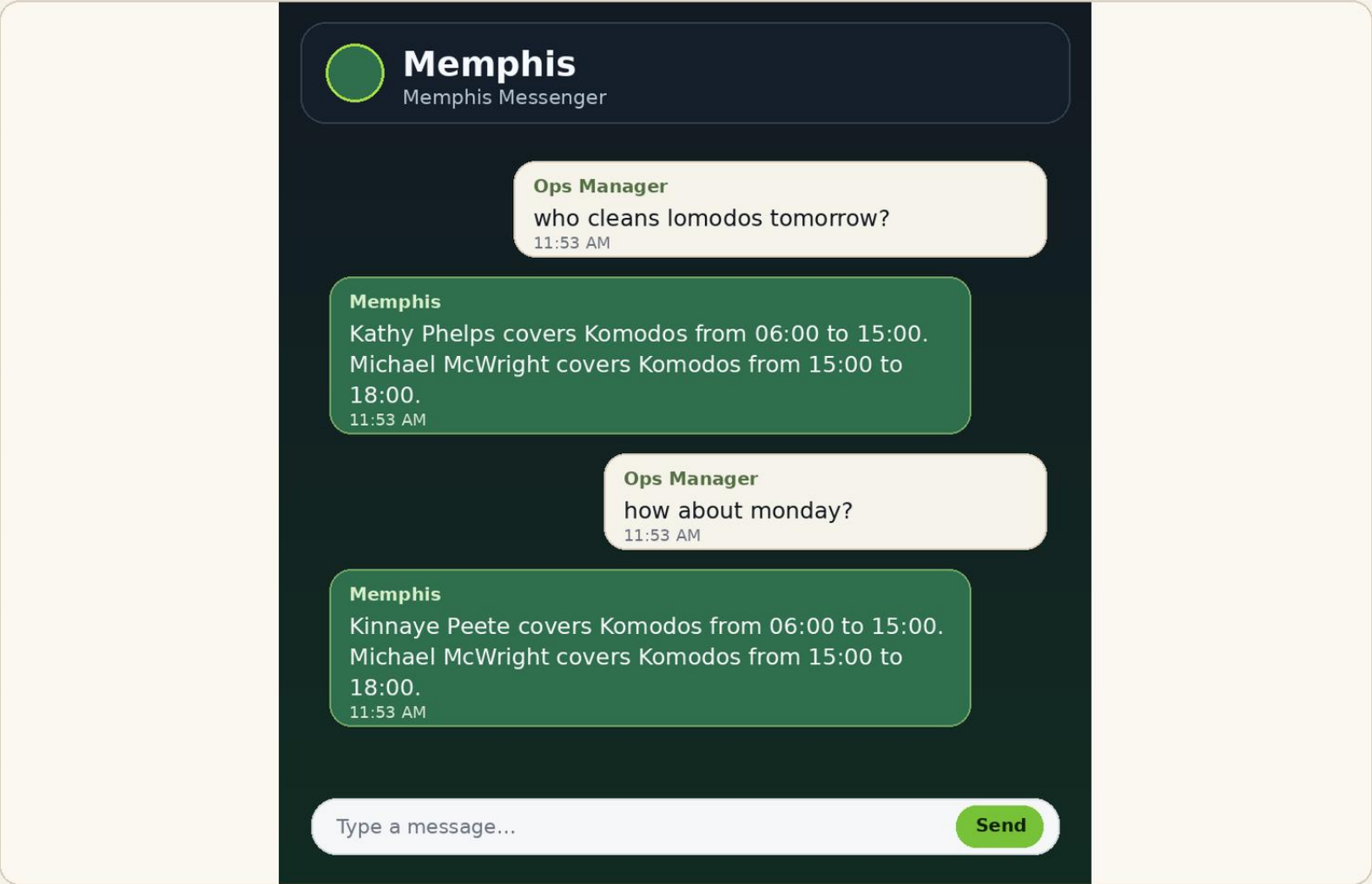
LOCATION	GROUP CODES	DIFFICULTY	PRIORITY	WEIGHT	NOTES	STATE	SAVE
Aquarium AQUX • exhibit	AQUARIUM	5	3	15	Optional workload notes	Manual settings	Save
Bonobos Men's Restroom BONM • restroom	BONOBOS_RESTROOMS	8	10	80	Optional workload notes	Manual settings	Save
Bonobos Women's Restroom BONW • restroom	BONOBOS_RESTROOMS	8	10	80	Optional workload notes	Manual settings	Save
Breezeway Men's Restroom BREM • restroom	BREEZEWAY_RESTROOMS	5	9	45	Optional workload notes	Manual settings	Save
Breezeway Women's Restroom BREW • restroom	BREEZEWAY_RESTROOMS	5	9	45	Optional workload notes	Manual settings	Save
Cathouse Cafe Men's Restroom CATM • restroom	CATHOUSE_CAFE_RESTROOMS	7	10	70	Optional workload notes	Manual settings	Save
Cathouse Cafe Women's Restroom CATW • restroom	CATHOUSE_CAFE_RESTROOMS	7	10	70	Optional workload notes	Manual settings	Save
China CHNX • exhibit	CHINA	4	4	16	Optional workload notes	Manual settings	Save

schedule build release-2026.04.30.2

What this screen shows

Daily ownership, week readiness, absence coverage, close-time settings, and advanced difficulty / priority weighting

PC view showing notifications and Memphis AI answers in one Messenger conversation.

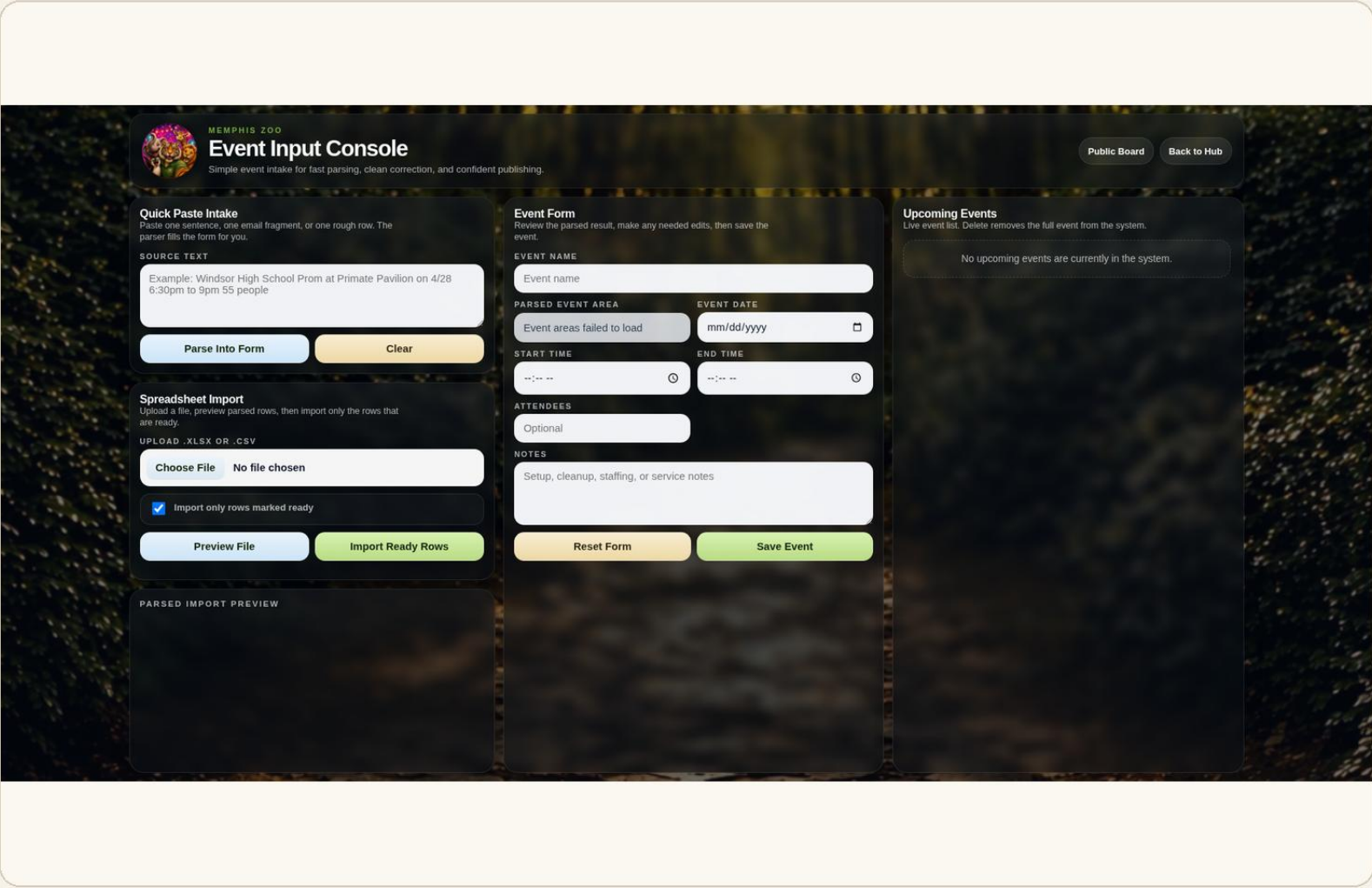


WHAT THIS SCREEN SHOWS

- Messenger is where work prompts and follow-up messages appear.
- Guest issues, area events, due-soon work, and overdue work can notify the right people the same way.
- Memphis AI can answer practical questions about staffing, schedules, coverage, tickets, scans, events, attendance, and contacts.
- The example also shows typo handling: “lomodos” is understood as Komodos.

Event Input Console

PC view for adding events by pasting, spreadsheet-style import, or a form.

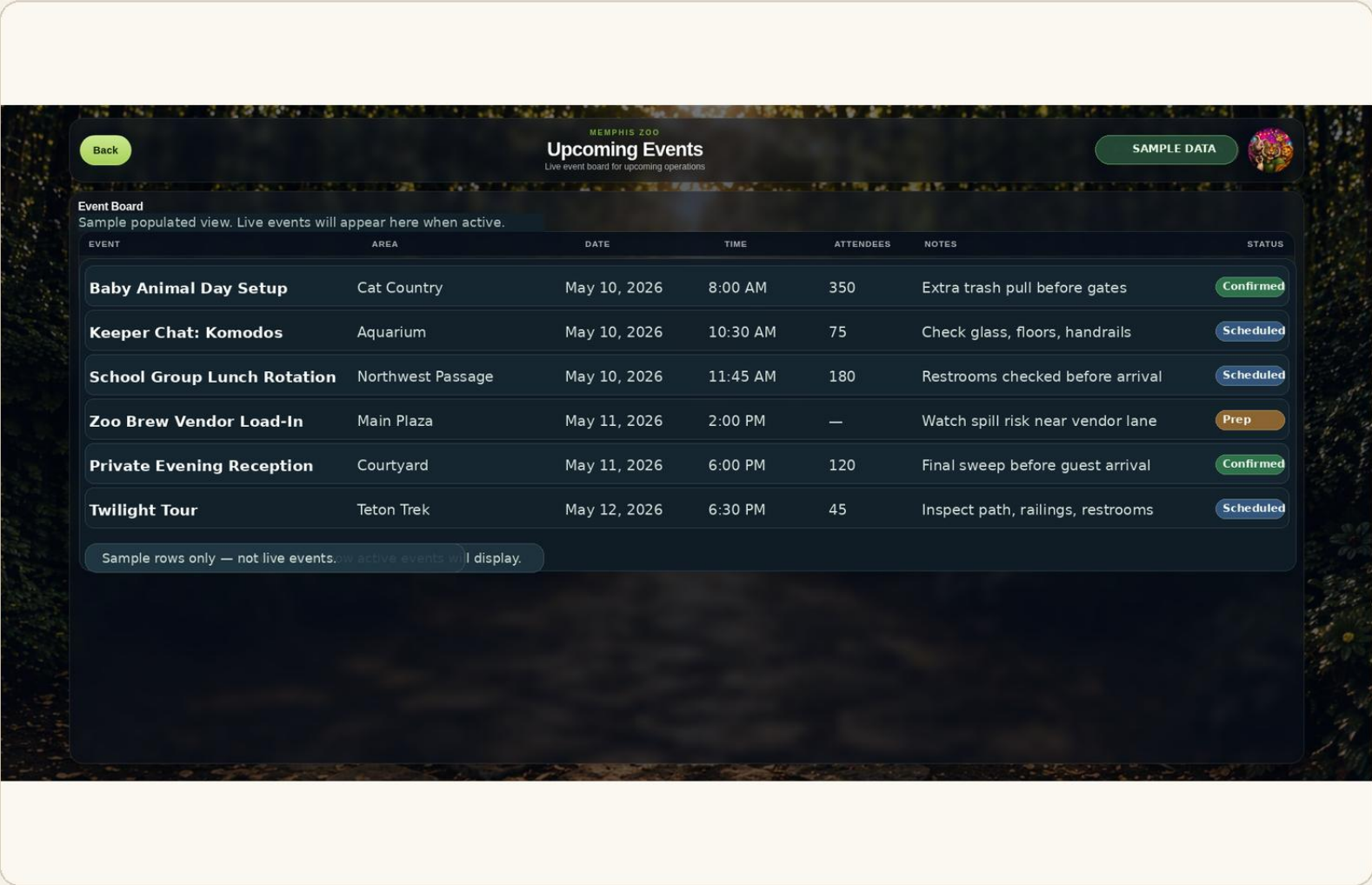


WHAT THIS SCREEN SHOWS

- Staff can paste event details, import spreadsheet-style information, or enter events by hand. The console pulls out key details such as event name, date, time, location, area, and notes.
- The area matters because custodians can be notified when an event affects their zone.
- Extra columns can be ignored, so staff do not have to clean up every spreadsheet before using it.
- This supports practical Messenger prompts, such as “events in your area today.”

Upcoming Events Display

PC view showing how active events appear.

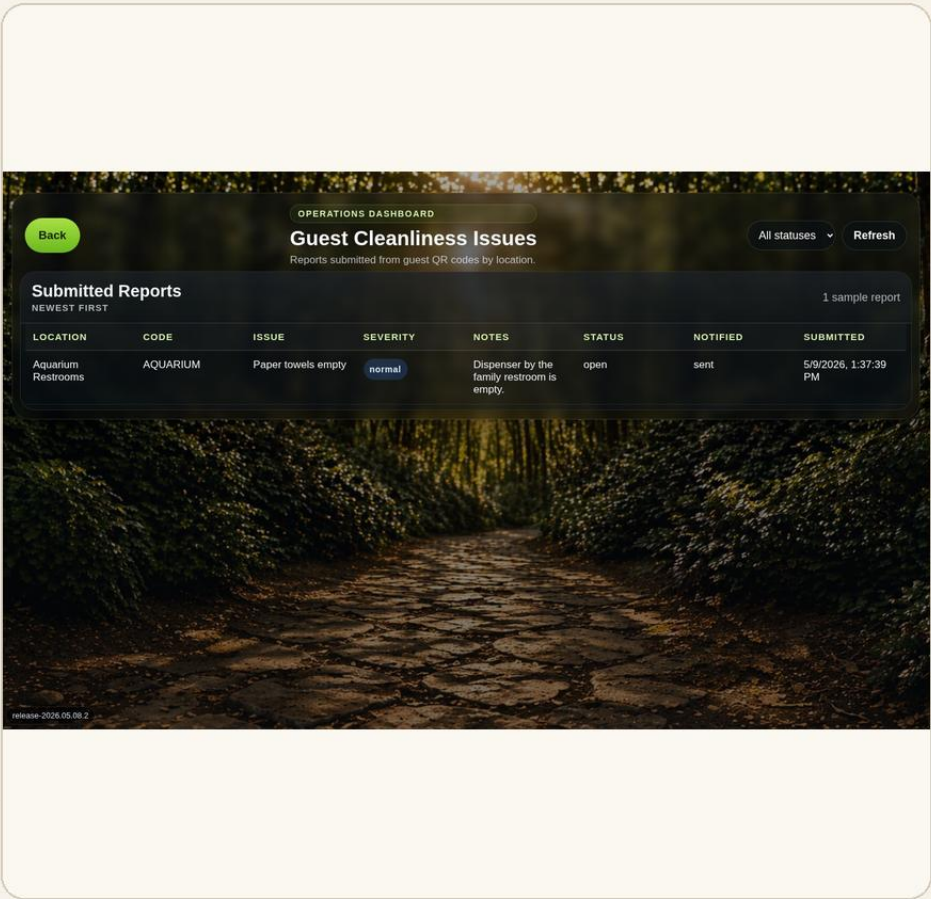
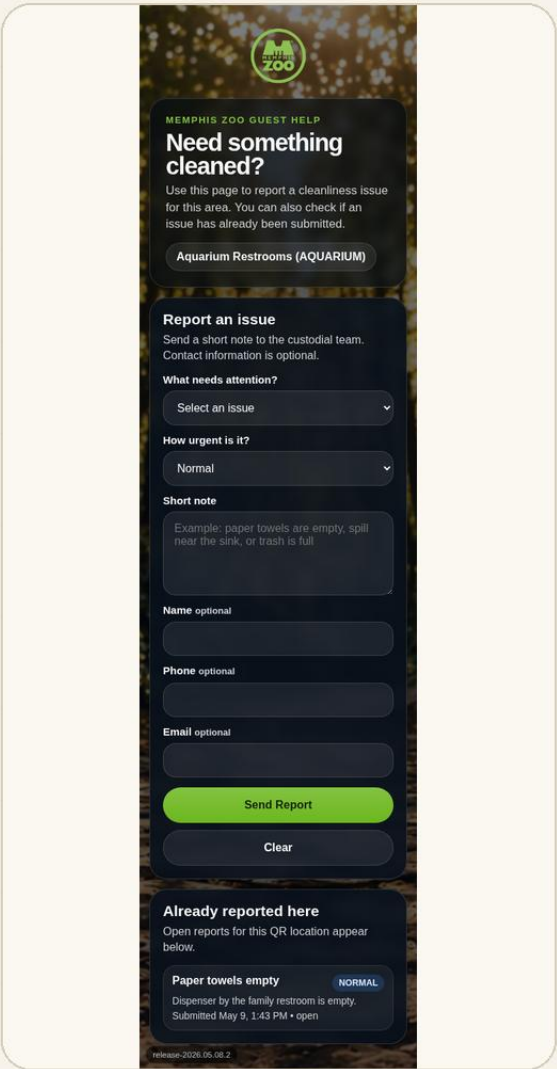


WHAT THIS SCREEN SHOWS

- Shows sample event rows so readers know what the screen looks like when events are active.
- When connected, the display refreshes as event records change.
- Events can be tied to areas so the right custodians get the right prompt.
- The sample rows show how dates, times, areas, notes, and status appear.

Guest Issue Reporting

A guest report should go straight to the right workflow, not sit as a loose comment someone has to route later.



Operations-side guest issue view

Guest-facing QR landing page with sample issue

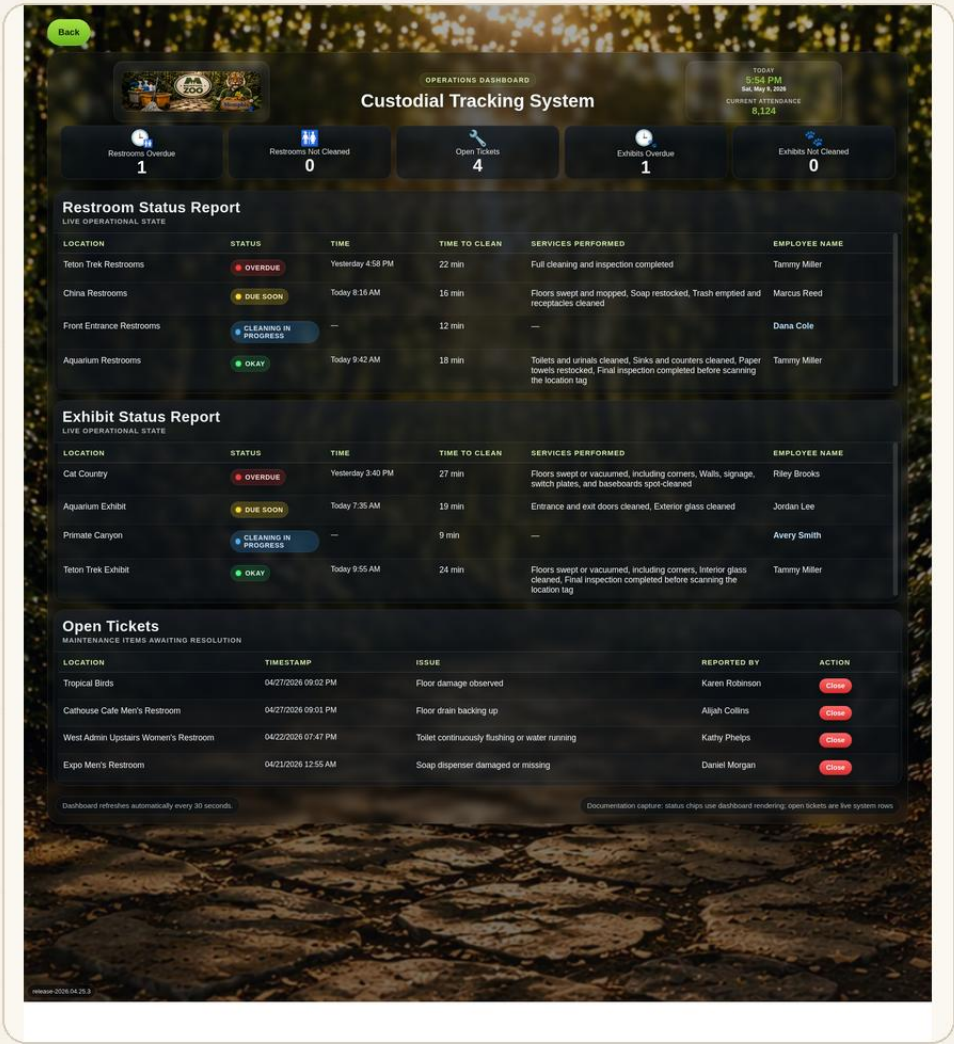
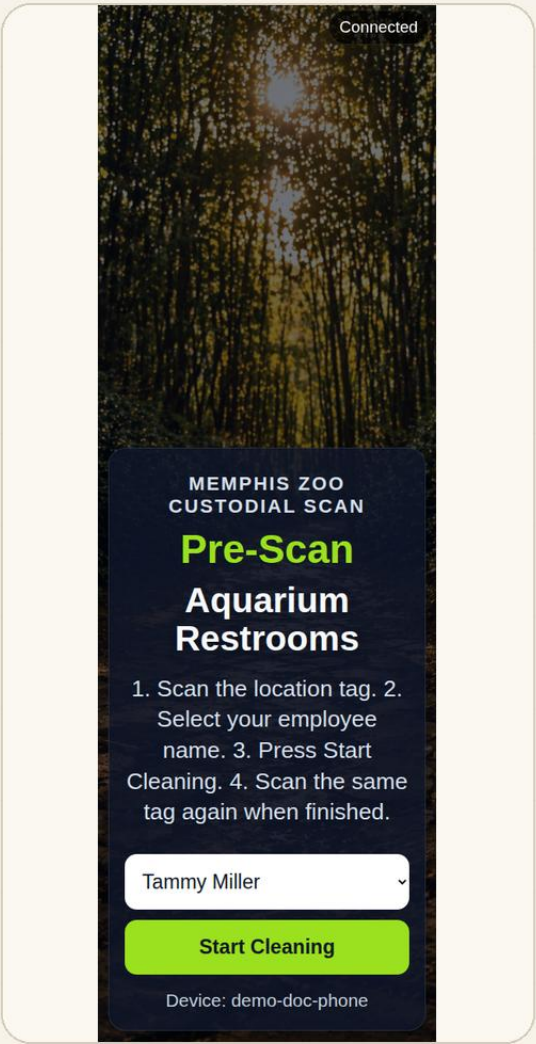
NOTIFICATION PATH

- A guest scans the location QR code and sees a simple report page for that area.
- The same sample issue appears on the guest page and the operations screen, making the path easy to follow.
- Messenger can notify the assigned custodian and operations contact, just like event, due-soon, and overdue notices.
- Managers can review issues from the operations screen and close them after follow-up.

Why this matters

Guest QR reporting turns a complaint into a task with a location, time, and follow-up path.

Scans guide custodians in the field and show managers what is okay, due soon, or overdue.



Custodian scan instructions

Manager / ops dashboard with realistic examples

WHAT THE WORKFLOW PROVES

- Custodians scan the location tag, select their name, start cleaning, and scan again when finished.
- The dashboard turns scans into simple status: OKAY, DUE SOON, OVERDUE, and open tickets.
- Restroom and exhibit examples appear together so managers can see the full custodial picture from a PC.
- This does not replace inspection; it tells managers when and where cleaned areas are ready to inspect.

Operational value

The phone guides the field work. The PC view shows what is ready to inspect and what needs follow-up.



MEMPHIS ZOO
CUSTODIAL SCAN

Restroom Completion Form

Aquarium Restrooms

Tammy Miller

Select at least one service performed before submitting.

Services Performed

☐ Toilets and urinals cleaned

☐ Sinks and counters cleaned

☐ Floor-to-ceiling deep cleaning completed, including corners, walls, cobwebs, vents, and fixtures

☐ Walls and partitions spot-cleaned for splashes, smudges, and fingerprints

☐ Trash emptied and receptacles cleaned, including feminine hygiene receptacles

☐ Floors swept and mopped, including corners and areas behind fixtures

☐ Mirrors and stainless steel surfaces cleaned and polished

☐ Toilet tissue, paper towel, and soap dispensers cleaned

☐ Hand dryer cleaned

☐ Baby changing station

☐ Toilet tissue, paper towel, and soap dispensers cleaned

☐ Hand dryer cleaned

☐ Baby changing station cleaned

☐ Toilet tissue restocked

☐ Paper towels restocked

☐ Soap restocked

☐ Air freshener applied

☐ Final inspection completed before scanning the location tag

☐ Full cleaning and inspection completed

Other Service Performed

Maintenance Issues Found

☐ Toilet leaking

☐ Toilet clogged or draining slowly

☐ Toilet continuously flushing or water running

☐ Toilet flush valve not functioning

☐ Urinal leaking

☐ Urinal clogged or draining slowly

☐ Urinal continuously flushing or water running

☐ Urinal flush valve not functioning

☐ Sink leaking

☐ Sink clogged or draining slowly

☐ Sink leaking

☐ Sink clogged or draining slowly

☐ Faucet loose or damaged

☐ Stall partition, door, lock, or coat hook damaged or missing

☐ Trash receptacle damaged or missing, including feminine hygiene receptacles

☐ Toilet paper dispenser damaged or missing

☐ Paper towel dispenser damaged or missing

☐ Soap dispenser damaged or missing

☐ Hand dryer not functioning

☐ Floor drain backing up

☐ Other maintenance issue found

Out of Order Status

☒ Yes

☐ No

Which stall(s), urinal(s), and/or sink(s) were out of order?

Notes

Submit Completion

Session ID: demo-restroom-001

Notes and submit

- Restroom checklist includes toilets/urinals, sinks/counters, mirrors, floors, trash, paper products, soap, odors, and final inspection.
- The same form captures fixture-level maintenance issues, out-of-order needs, and notes before submission.
- The final submission creates the record managers use for follow-up and inspection timing.

Notes and submit

- Exhibit checklist includes floors, glass, stairs, railings, doors, signage, baseboards, cobwebs, trash, and final inspection.
- Maintenance issues and notes are captured before submission, just like the restroom workflow.
- The final submission creates the record managers use for follow-up and inspection timing.

Actual Project Cost and Monthly Cost13

The cost is based on the attached equipment table. Phones are a one-time equipment purchase; there is no carrier plan and no current monthly software cost.

ACTUAL EQUIPMENT PURCHASES

Item	Qty	Unit Cost	Subtotal
Moto G Phones	10	\$39.99	\$399.90
USB Charging Cords	2	\$7.59	\$15.18
Charging Station	1	\$31.98	\$31.98
Phone Holsters	10	\$12.99	\$129.90
NFC Tags	4	\$13.99	\$55.96
NFC Reader	1	\$37.99	\$37.99

Total Project Cost	\$670.91
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Monthly cost clarification

The equipment above is the real implementation purchase total. The phones do not require a carrier subscription because they run on zoo Wi-Fi. Current monthly software cost remains \$0/month for the frontend/backend services

CURRENT SERVICE COST

GitHub Pages	\$0/month current	Hosts the static frontend pages.
Supabase	\$0/month current	Current database service remains on the free tier unless scale, backups, or operating needs change.
Render	\$0/month current	Current hosting remains on the free tier unless scale or operating needs change.

EXPANSION MODEL

- Adding locations mainly means adding NFC/QR points and setup records, not buying a new software package.
- Adding phones means low-cost hardware and accessories, not a carrier bill when phones stay on zoo Wi-Fi.
- Ongoing costs are mostly replacements: phones, NFC tags, charging gear, or optional cloud upgrades.
- The system avoids paying for broad CMMS features the custodial team may not need right now.

Most commercial systems cover pieces of this. Matching the full workflow usually means multiple products or higher-tier plans.

MaintainX

Monthly: \$20-\$75 per user on published paid tiers.

Setup/equipment: implementation may be separate. Phones/tablets, labels, and kiosk setup are usually handled by the customer. Strong CMMS, but it does not provide this full zoo-specific workflow out of the box.

OrangeQC

Monthly: about \$250-\$500, plus add-ons and extra inspectors.

Setup/equipment: QR shortcuts are available, but field devices, printed codes/tags, and rollout work are still separate. Good inspection/ticket fit, but not a full custom operations hub.

UpKeep

Monthly: about \$24-\$55 per user on published tiers.

Setup/equipment: published implementation/training packages range from about \$500 to \$5,000; field devices/labels are still separate. Broad platform that may be more than custodial teams need.

Why the monthly cost jumps

Commercial tools can become expensive once paid users, QR workflows, offline mode, analytics, setup help, portals, APIs, or extra modules are added.

Initial setup difference

Many subscriptions do not include phones, tablets, QR labels, NFC tags, kiosk setup, data cleanup, or rollout work. Those are usually startup costs.

Most commercial systems cover pieces of this. Matching the full workflow usually means multiple products or higher-tier plans.

Limble CMMS **Monthly: quote/calculator model on public page.**

Setup/equipment: QR/work request features exist, but pricing and rollout costs depend on quote. Devices and labels are typically separate. Strong CMMS, broader than this use case.

SafetyCulture **Monthly: team inspection/checklist platform pricing.**

Setup/equipment: mobile devices, printed QR codes, and rollout configuration are outside the base software concept. Good inspections, but not schedule + Messenger + events + Memphis AI.

Is there one system this comprehensive?

Public vendor products cover important pieces, but one ready-made tool with scans, guest QR issues, events, scheduling, Messenger, reminders, and Memphis AI is not obvious.

Cost conclusion

The custom path wins on cost: current monthly software cost is \$0, startup hardware is low, and future spending is mostly replacements or optional upgrades.

Comparable venues already use QR reports, smart restrooms, mobile inspections, guest feedback, and dashboards. The Memphis Zoo value is combining those ideas into one low-cost custodial workflow.

COMPARABLE VENUE EXAMPLES

- Airports: ELERTS describes restroom QR reporting where guests scan a code, submit details or a photo, and send a location-tagged report to maintenance.
- Airports: TRAX describes restroom programs using guest feedback, connected dispensers, cleaning alerts, mobile workflows, and performance data.
- Large venues: TRAX markets restroom intelligence for airports, arenas, and stadiums — high-traffic facilities that need fast cleaning visibility.
- Hotels: Bitly and The QR Code Generator describe QR housekeeping requests that reduce calls, route guest needs, and give staff a clearer digital queue.
- Visitor attractions: theme parks and aquariums face the same challenge as a zoo: spread-out restrooms, high guest volume, event spikes, and location-specific team prompts.

WHAT THIS SUPPORTS

- The individual ideas are already used in real facilities: QR intake, mobile checklists, guest feedback, smart restroom data, dashboards, and alerts.
- This system adapts those ideas to zoo custodial work without buying an oversized enterprise platform.
- The advantage is the combination: schedules, NFC/QR proof, guest reports, area events, reminders, Messenger, manager visibility, and Memphis AI in one place.

SAFE COMPARISON

Simple claim: other facilities validate the direction. This version is tailored to the zoo’s daily operation at the current \$0/month software cost.

SOURCES CHECKED

ELERTS airport restroom QR reporting; TRAX Kansas City International Airport case study and TRAX facility categories; Janitorial Manager public scan QR codes; Bitly hotel housekeeping QR requests; The QR Code Generator hotel housekeeping QR requests; vendor examples for smart washroom use in high-traffic venues.